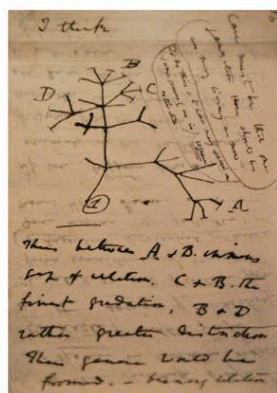


TREE OF LIFE

Using a branching tree as a way of showing the diversity of life was first suggested by the German naturalist Peter Pallas in 1766. Since then many such trees have been constructed. Initially tree-like, complete with bark and leaves, they later became more diagrammatic, and took account of evolutionary theories. Modern, computer-generated trees of life present many different ideas of how living organisms are related.



DARWIN'S FIRST TREE

The first person to produce a tree of life that reflected the concept of evolution was Charles Darwin. He sketched the first of his ten evolutionary trees of life in 1837. It was a simple branching diagram, which he developed further before publishing it in the *Origin of Species* in 1859. The lettered branches show how he thought his theory might work – the more branch points separating an organism from its ancestor (numbered 1), the more different the organism will be. In 1879, Ernst Haeckel took the idea further, with a tree that showed animals evolving from single-celled organisms.

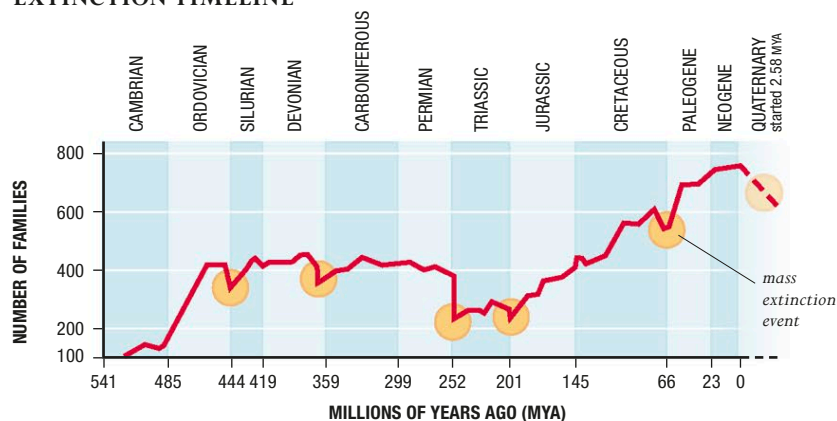
Today, DNA and protein analyses as well as morphology are used to construct evolutionary trees and establish the genetic relationships between organisms. Vast data sets require computers to generate the trees, which are continually refined as new species and information are discovered.

Traditionally, trees of life emphasised vertebrate groups and relationships as those of the microscopic archaea and bacteria, and protoctists (those eukaryotes not classified as plants, animals, or fungi) were more problematic. However, modern genomic studies have revealed an extraordinary diversification and complexity of relationships within the bacteria and within the early, or basal, eukaryote groups.

MASS EXTINCTIONS

Mapping all life forms that have ever existed on a tree is difficult as, over time, more than 95 per cent of all species have become extinct. A mass extinction occurs when a large number of species dies off at the same time. This has happened five times in the past. The best-known extinction, which wiped out the dinosaurs, occurred at the end of the Cretaceous Period; it is thought to have been caused by a meteor impact combined with volcanic activity. As habitats are rapidly destroyed by human activity, it is likely that there will be another extinction event in the future.

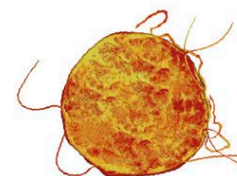
EXTINCTION TIMELINE



READING THE TREE

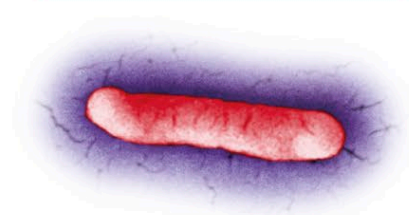
This diagram shows how life evolved from simple organisms, such as the Archaea – which appeared about 3.4 billion years ago – to complex life forms, such as animals, which appeared about 650 million years ago. It also shows the diversity of the vertebrates (see pp.34–35), which have a disproportionate representation here. The circles indicate points where two or more groups of organisms have branched from a common ancestor at about the same time. Only extant species are shown, so extinct groups, such as the reptilian dinosaurs, are missing.

LIFE BEGINS



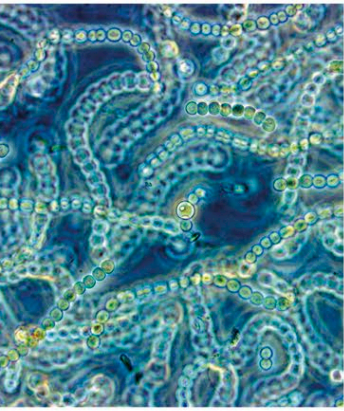
ARCHAEA

BACTERIA



STRUCTURE OF LIFE

All forms of life belong to one of three domains: Archaea, Bacteria, and Eukaryota. Archaea and bacteria are unicellular and lack a cell nucleus – they used to be classified together as Prokaryota. Eukaryotic organisms tend to be multicellular and each cell contains a nucleus, within which DNA is stored. This table shows the three domains and the four eukaryote kingdoms. Despite appearances, the Archaea and Bacteria are the largest groups – although only about 20,000 species have been described, estimates exceed 4 million species. Among eukaryotes, the groups that make up the protoctists and invertebrates are far more numerous in terms of species than vertebrate groups.

ARCHAEA	EUKARYOTES
BACTERIA	PROTOCTISTS
	PLANTS
	MOSSES
	LIVERWORTS
	HORNWORTS
	LYCOPHYTES
	FERNS AND RELATIVES
	CYCADS, GINKGOS, AND
	GNETOPHYTES
	CONIFERS
	FLOWERING PLANTS
	FUNGI
	MUSHROOMS
	SAC FUNGI
	LICHENS
	ANIMALS
	INVERTEBRATES
	CHORDATES

CYANOBACTERIA